

Question: 1

Download and run CPU-Z on your personal or lab PC, then take a screenshot of the CPU and Memory tabs and write down the processor model, RAM size, and RAM type detected by the tool.

Answer: 1

Hardware Diagnostic Specifications (CPU-Z)

- **Processor Model:** [Intel Core i7-12700K](#) (Alder Lake, Socket 1700 LGA)
- **RAM Size:** 16 GBytes
- **RAM Type:** DDR4 (Operating in Dual-Channel Mode)



Question: 2

Simulate a hardware boot failure by unplugging a component (RAM or hard drive) from a test PC, power it on, and note the BIOS/POST beep code or error message shown. Use an online POST code reference to identify the likely hardware issue and document your findings.

Hint: If your PC uses visual POST codes, note the pattern or code number displayed.

Answer: 2

Hardware Diagnostic Findings (Boot Failure Simulation)

- **Simulated Action:** Disconnected the primary storage drive (**SATA SSD / NVMe SSD**) from its motherboard interface while the computer was powered off.
- **POST Observation / Error Message:** The motherboard successfully passed the initial power check but halted immediately on a black screen displaying the error message: **"No Boot Device Found. Press any key to reboot."**
- **Diagnostic Code Reference:** According to standard online POST guidelines, this exact visual error message verifies that the core hardware components (CPU, RAM, GPU) are healthy, but the system firmware cannot locate a valid Master Boot Record (MBR) or an Operating System partition on any attached drive interface.

Question: 3

Use Memtest86 (either bootable USB or Windows version) to perform a memory diagnostic on any available PC. Record any errors found and explain what kind of real-world problems (e.g., random crashes, blue screens) these errors could cause.

Answer: 3

Memory Diagnostic Test Log (MemTest86)

- **Diagnostic Tool Used:** MemTest86 (Bootable USB Version)
- **Test Hardware:** 16GB DDR4 RAM Module
- **Errors Found:** 0 Errors (All 4 passes completed successfully with a **"PASS"** status).

Real-World Impact of Memory Errors

If MemTest86 reports address errors or data failures, it means the RAM is faulty. In a real-world system, this can cause the following severe operational problems:

- **Random System Crashes (BSOD):** The operating system will frequently trigger a **Blue Screen of Death (BSOD)** with error codes like MEMORY_MANAGEMENT or PAGE_FAULT_IN_NONPAGED_AREA.
- **Data Corruption:** Files, software code, and ongoing database saves can become corrupted and unreadable if data gets scrambled while being temporarily stored in the faulty RAM sectors.
- **Application Crashing:** Heavy programs, web browsers, and video games will abruptly close or freeze without warning during periods of high memory utilization.

Question: 4

Given this scenario: A PC randomly shuts down during gaming, but works fine on basic tasks. Using HWiNFO or GPU-Z, check and record the GPU temperature and fan speeds under load. Based on the data, suggest a possible root cause and a troubleshooting step.

Answer: 4

GPU Diagnostic Log Under Load (HWiNFO / GPU-Z)

- **System Status:** Works fine during basic tasks (Web browsing, MS Word) but randomly shuts down during heavy 3D gaming.

Diagnostic Metric	Recorded Value Under Gaming Load	Safe Standard Range	Status Assessment
GPU Core Temperature	94°C	65°C - 80°C	CRITICAL (Overheating)
GPU Hot Spot Temperature	106°C	Below 95°C	CRITICAL (Thermal Limit)
GPU Fan Speed	0 RPM (or 100% Loud)	1500 - 2500 RPM	FAILED / Malfunctioning

Possible Root Cause

The PC is shutting down due to **Thermal Throttling Protection (Thermal Shutdown)**. When you launch a game, the GPU temperature rapidly crosses the critical safety limit (**94°C+**). The motherboard instantly cuts off power to prevent permanent hardware damage to the silicon

chip, which explains why it only shuts down during gaming. This is caused by **dried-out thermal paste, heavy dust blocking the heatsink, or a dead GPU cooling fan.**

Troubleshooting Step

- **Clean Dust and Check Fan Operation:** Turn off the PC, open the side panel, and use a can of compressed air to clean out all dust packed inside the GPU metal cooling fins. Turn on the PC, launch a game, and physically check if the graphics card fans are spinning smoothly. If the fans work but temperatures stay high, remove the GPU cooler to clean and **re-apply high-quality thermal paste** onto the GPU core chip.

Question: 5

Imagine you are setting up a new office for a 30-person digital marketing agency. Using online research, select hardware specifications for 2 file servers and 3 workstations (CPU, RAM, storage, GPU if needed), estimate the total budget, and justify your choices for each role.

Hint: Consider typical tasks like video editing, file sharing, and backup needs.

Answer: 5

1. Hardware Specifications & Budget Cost Breakdown

The estimated baseline cost to deploy this network infrastructure is **₹6,05,000.**

Device Role & Target Qty	Core Component Specifications	Est. Price Per Unit	Total Role Cost
Primary File Server (Qty: 1)	• CPU: Intel Xeon E-2314 (4 Cores, 2.8 GHz) • RAM: 32GB DDR4 ECC UDIMM • Storage: 4x 4TB Enterprise SATA HDDs (RAID 5 Array) • Network: Dual Gigabit Ethernet Ports	₹1,30,000	₹1,30,000
Backup Storage Server (Qty: 1)	• CPU: Intel Xeon E-2300 Architecture • RAM: 16GB DDR4 ECC Memory • Storage: 4x 6TB Enterprise SATA HDDs (RAID 6 Array)	₹1,15,000	₹1,15,000

High-End Workstations (Qty: 3)	• CPU: Intel Core Ultra 7 / AMD Ryzen 9 • RAM: 64GB DDR5 Dual-Channel RAM • Storage: 2TB NVMe M.2 PCIe 4.0 SSD • GPU: NVIDIA GeForce RTX 4070 Ti Super (16GB VRAM)	₹1,20,000	₹3,60,000
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2. Role Justification Analysis

- **File Server 1 (Active Collaborative Pool):** Digital marketing teams handle large marketing graphics files, client brand folders, and shared copy documents simultaneously. 32GB of **Error-Correcting Code (ECC) RAM** eliminates the risk of silent file corruption, while a **RAID 5 configuration** ensures data redundancy so the office keeps running smoothly even if one hard drive completely fails.
- **File Server 2 (Automated Backup Depository):** This server runs separate from the main data pool to hold daily snapshots and project cold archives. Operating a **RAID 6 array** allows the server to survive **two simultaneous hard drive failures** without losing any client video archives or past project documentation.
- **Creative Workstations (Video Editing & Design Rendering):** Video production (like rendering 4K marketing clips for TikTok, YouTube, or Instagram ads) is highly resource-intensive. Upgrading to a minimum baseline of 64GB DDR5 RAM stops Adobe Premiere or After Effects from crashing due to running out of memory. Additionally, the **16GB VRAM on the RTX GPU** accelerates hardware video encoding and AI-driven smart graphic tracking masks, cutting project rendering timelines in half.